

Case Study 1: Ethical Failures Behind the Boeing 737 MAX Disasters

Ethical Theory Used: Deontology (Kantian Ethics/Utilitarianism Ethics)

The Boeing 737 MAX disasters in 2018 and 2019 underscore a profound failure of ethical responsibility in engineering and corporate leadership. From a deontological perspective, which emphasizes duty, honesty, and respect for human life regardless of consequences, Boeing's actions were unethical. The crashes were not a technical failure but the result of deliberate choices that violated fundamental moral and professional duties.

One of the central ethical failures was Boeing's lack of transparency regarding the Maneuvering Characteristics Augmentation System (MCAS). This software system had the authority to override pilot control, yet the pilots were not informed of its existence or operation. From a Kantian perspective, the essential safety information contravenes the obligation of honesty regarding pilots and passengers; it is a means to achieve the objective of sustaining market competitiveness with Airbus, rather than an end in itself. Deontological ethics rejects any justification that relies on beneficial outcomes if it requires deception or the neglect of moral duty.

Additionally, Boeing minimized pilot training requirements to reduce costs and accelerate regulatory approval. Internal communications revealed that engineers raised safety concerns, yet these warnings were ignored or overridden by management. Engineers have a professional duty, reinforced by engineering codes of ethics, to prioritize public safety above all else. By allowing economic pressures to supersede safety obligations, Boeing failed to fulfill its duty. Even if no accidents had occurred, the decision-making process itself would still be unethical under deontological ethics because it violated fundamental moral rules.

From a utilitarian perspective, Boeing's actions might initially appear defensible if they were believed to maximize overall benefits, such as economic efficiency, job preservation, and faster delivery of aircraft to meet global demand. However, utilitarianism ultimately condemns Boeing's decisions once the consequences are fully considered. The loss of hundreds of lives, long-term psychological trauma, and erosion of public trust in aviation far outweigh any short-term economic gains. Rather than maximizing overall well-being, Boeing's actions resulted in widespread harm, demonstrating that even under a consequence-based ethical framework, their conduct was unjustifiable.

Responsibility for these failures lies primarily with Boeing's corporate leadership, which knowingly made decisions that compromised safety. However, responsibility is not limited to executives alone. Engineers who remained silent or complied with unethical directives share partial moral responsibility, particularly if they were aware of the risks involved. The Federal Aviation Administration (FAA) also bears responsibility due to its reliance on Boeing's self-certification process, reflecting a significant breakdown in regulatory oversight.

In conclusion, the Boeing 737 MAX case shows how ethical failures in engineering can stem from both flawed decision-making and organizational culture. Under deontological ethics, Boeing violated its duty to protect human life and act with honesty, while utilitarianism reveals that its actions failed to produce the greatest good for society. This case illustrates that ethical engineering requires more than technical expertise; it demands moral courage, accountability, and a commitment to prioritizing human safety over profit. When these ethical responsibilities are ignored, the consequences can be catastrophic, as evidenced by the tragic loss of hundreds of lives.

WORD COUNT: 499

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Case Study 2: The Ethics of Driverless Cars

Ethical Theory Used: Utilitarianism

The development of autonomous vehicles introduces complex ethical challenges, particularly when self-driving cars are required to make life-and-death decisions. Utilitarianism, which evaluates actions based on their consequences and aims to maximize overall well-being, is often proposed as the most practical ethical framework for making these decisions. However, while utilitarianism offers clarity and structure, it also raises significant moral concerns.

Self-driving cars may face scenarios similar to the classic trolley problem, where harm is unavoidable, and the system must choose between two bad outcomes. Suppose a vehicle may need to decide whether to protect its passenger or avoid hitting multiple people. From a utilitarian perspective, the ethically correct action is the one that minimizes total harm, even if it results in the death of a single individual. This approach aligns with the broader goal of reducing traffic fatalities and improving overall road safety.

Utilitarianism is beneficial in this context because it can be translated into algorithms and data-driven decision-making. It allows engineers to quantify outcomes and design systems that statistically save the most lives over time. Given that autonomous vehicles are expected to reduce accidents caused by human error, a utilitarian framework supports their deployment, despite occasionally requiring morally difficult decisions.

However, this approach has ethical problems. One major concern is that moral values are not universal. Studies show that people from different cultures disagree on what outcomes are morally acceptable, making it difficult to define a single "correct" algorithmic response. Additionally, many individuals are uncomfortable with the idea that a machine could decide to sacrifice them for the greater good, especially without their consent.

Critics also argue that utilitarianism can justify morally troubling actions. From a deontological perspective, intentionally sacrificing an innocent person is always wrong, regardless of the benefits. Furthermore, questions of responsibility remain unresolved. If an autonomous vehicle causes harm, it is unclear whether blame lies with the programmer, manufacturer, owner, or regulator.

In conclusion, while utilitarianism offers a practical framework for addressing ethical decision-making in autonomous vehicles, it falls short in resolving issues related to consent, responsibility, and moral rights. The ethics of driverless cars reveal that moral decision-making cannot be fully objective or universal. As autonomous technology evolves, ethical considerations must encompass engineers, policymakers, and the public to ensure these systems conform to societal values.

WORD COUNT: 380

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